Pinealon

0.5–1 mg daily SC or IN for 10–20 days, 0.1 – 0.2 mg IN continuously

Pinealon is an experimental tripeptide proposed for neuroprotection, preservation of learning and memory, and cognitive recovery after biological stress. [1–11] Pinealon (EDR) alters oxidative-stress responses, ERK signaling, and cell-cycle behavior in experimental cells, and preserved learning or neuronal structure in selected animal models. [1–7] Intracellular gene regulation is a proposed explanation for these effects; DNA interactions have been studied biophysically, while transporter-mediated entry and downstream gene regulation remain context-dependent mechanisms. [8,12,13,22] Human cognitive benefits are plausible on the combined preclinical and limited clinical evidence. [10,11,16,20]

💉 Short-term tolerability is plausible under studied conditions, but longer-term safety is unestablished. [16,17,19,23]

⭐ Learning and neuronal-structure benefits have been observed in specific rat and mouse models. Comparable human benefits are plausible. [2–7]

⭐ Cognitive and post-traumatic benefits are possible based on limited human reports and compatible preclinical findings. [10,11,16,20]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

A patent reports reassuring general, blood, organ, hepatic, renal, cardiovascular, and respiratory observations in repeated-dose animal studies lasting up to six months. These are patent-reported findings with limited independently assessable detail. [16] An eight-week mouse preprint reported no detected adverse effects under its study conditions. [19] Limited human reports provide some support for short-term tolerability. [16,23]

How Evidence Scope Was Applied in This Review

Evidence was evaluated here with an adjustment for source quality. Much of the Pinealon evidence comes from a single regional source, including small or methodologically limited studies that have not been independently replicated. When a claim depended primarily on this evidence, its evidence classification was reduced to reflect the additional uncertainty while still recognizing the evidence as informative.

Evidence Overview

The strongest Pinealon evidence is preclinical. It reduced stress-induced ROS in rat neuronal and neutrophil preparations, reduced necrotic death in PC12 cells, and altered ERK1/2 activation and cell-cycle distribution. [1] Rat studies reported learning or retention benefits in prenatal

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

hyperhomocysteinemia, hypoxic stress, and experimental diabetes. [2–5] Amyloid-model studies reported preservation of dendritic spines; the mouse study's long-term potentiation results were trends rather than statistically established functional recovery. [6,7] In induced neurons derived from three older female donors, EDR increased primary processes and total dendritic length; the oxidative-DNA-damage reduction was a nonsignificant. [9] A 2026 mouse preprint reported a working-memory trend towards improvement over eight weeks. [19]

Human reports describe occupational adaptation, psychoemotional symptoms, biological-age measures, and persistent symptoms after brain injury. [10,11,16,17,20,23] An oral Pinealon report in locomotive workers described improved adaptation-related measures after two weeks. [20] The truck-driver study reported its best effects with combined Pinealon and Vesugen. [11] The brain-injury patent describes a small randomized saline-controlled IM comparison, but its outcome percentages do not reconcile with its stated sample size. [16] These findings support possible cognitive and post-traumatic benefits, with substantial uncertainty about their magnitude and attribution.

Community Applications and Experience

Community applications summarized here include memory, focus, brain fog, concussion recovery, cognitive healthspan, mental stamina, mood stability, sleep quality, vivid dreaming, and circadian adjustment. The reported experiences are mixed and anecdotal, including clearer thinking, better sleep, fragmented sleep, stimulation, fatigue, or no noticeable effect. These applications and experiences are speculative; they are not measures of treatment efficacy or tolerability.

Dosing Patterns

There is no established human PK or SC or intranasal regimen. Published oral use includes a two-week occupational report. [16,20]

Published Research Exposure

Animal Regimen	Human Equivalent Dose (150 lbs)
rat IP — prenatal neuroprotection [2]	0.11 mg daily for 5 days
mouse IP — dendritic-spine preservation [7]	2.2 mg daily for 2 months

- In the diabetic-rat study, Pinealon preserved aspects of previously learned behavior and altered hippocampal NMDA-receptor subunit gene expression. [5]
- In the mouse Alzheimer model, EDR increased dendritic-spine density, with sex-dependent effects on mushroom-shaped spines. The publication subsequently corrected duplicated male/female illustrative images; the authors stated that the conclusions were unaffected. [7,15]

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

Pinealon

0.5–1 mg daily SC or IN for 10–20 days, 0.1 – 0.2 mg IN continuously

Speculative Experimental Dosing

In community use two speculative, experimental daily cognitive-healthspan patterns emerge

- **0.5–1 mg daily SC or IN for 10–20 days**

Sometimes SC doses of up to 2 mg daily and courses totaling about 20 mg are discussed.

Alternatively, a small continuous intranasal dose is also discussed.

- **0.1-0.2 mg daily IN for continuously**

Benefits

Pinealon's most defensible benefits concern resistance to neuronal stress and preservation of learning or neuronal structure in experimental systems. [1–9] Broad nootropic enhancement, sleep normalization, and cognitive healthspan benefits in healthy people remain speculative.

- **Learning and memory:** Learning or retention improved in specific prenatal, hypoxic, and diabetic rat experiments. [2–5] Human cognitive benefit is possible, supported by the preclinical findings and limited occupational or post-traumatic reports. [10,11,16,20]
- **Neuronal oxidative protection:** Pinealon reduced stress-induced ROS in experimental rat cells and increased the resistance of offspring neurons after maternal treatment in a prenatal stress model. [1,2] Comparable protection in humans is plausible, but redox effects are context-dependent: a human report involving Pinealon and Vesugen described pro-oxidant activity. [17]
- **Synaptic structure preservation:** Dendritic-spine preservation was observed in amyloid-exposed mouse cultures and in the 5xFAD mouse model. [6,7] Increased dendritic processes and length in induced human neurons support a plausible human structural benefit. [9]
- **Post-traumatic cognition:** Improvement in memory, attention, headache burden, emotional balance, and performance after traumatic brain injury is possible based on limited clinical descriptions and compatible preclinical findings. [10,16]
- **Neuroinflammatory regulation:** Cytokine and caspase-related responses changed in aged rats exposed to acute hypoxia. [4] Broader regulation of neuroinflammation in humans is plausible; the measured markers do not independently demonstrate formation of new neurons.
- **Neuroplasticity support:** Changes in dendritic arborization, spine structure, NMDA-receptor gene expression, and ERK signaling make improved neuroplastic resilience plausible in humans. [1,5–9] These structural and molecular endpoints are distinct from demonstrated functional recovery.
- **Healthy brain aging:** Preservation of neuronal structure is plausible from the induced-neuron findings, but the oxidative-DNA-damage result was a nonsignificant trend, and mitochondrial, lysosomal, p16, and lamin B1 endpoints did not show significant improvement. [9]

© 2026 Researcher6076

Preservation of human cognitive function with aging and extension of healthspan remain speculative.

- **Mood and mental stamina:** Improvements in psychoemotional or adaptation measures are possible in occupational or post-traumatic contexts. [11,16,20] General mood or performance enhancement in healthy adults is speculative.
- **Sleep and circadian effects:** Better sleep continuity, altered dreaming, easier waking, and circadian adjustment are speculative community applications. Increased serotonin expression in aging brain-cortex cultures provides a possible neurochemical hypothesis, but does not determine the direction of a human sleep effect. [21]

Potential Applications

- **Traumatic brain injury** [10,16]
- **Metabolic cognitive impairment** [5]
- **Hypoxic brain stress, including stroke-related injury** [3,4]
- **Neurodegenerative disorders** [6,7,18]
- **Cognitive aging** [9,17,20,23]

Cautions

- **Pregnancy and breastfeeding** [2]
- **Children and adolescents** [2]
- **Neurologic conditions** Caution with unstable neurologic disease is a mechanistic precaution [1,5,21]
- **Psychiatric vulnerability** Precautions in mania, severe anxiety, psychosis, or other unstable psychiatric conditions are mechanistic [21]
- **Sleep disorders**
- **CNS-active combinations:** Additive or opposing effects with stimulants, sedatives, antidepressants, anticonvulsants, or other investigational neuroactive peptides are mechanistically plausible. [1,5,21]
- **Active malignancy:** EDR altered cell-cycle distribution in PC12 cells. This supports a mechanistic precaution in growth-sensitive disease. [1]

Speculative Adverse Effects Described in Community Use.

- **Headache**
- **Sleep disruption**
- **Stimulation or anxiety**
- **Fatigue**

Pinealon

0.5–1 mg daily SC or IN for 10–20 days, 0.1 – 0.2 mg IN continuously

Diminished Response and Cycling

Pinealon is not established as a receptor agonist. Its proposed intracellular actions do not rule out adaptation in signaling, gene expression, or neuronal function. [1,8,12,13] Cycling is speculative.

Biological Mechanism

The underlying molecular mechanisms of peptide bioregulators such as Pinealon remain poorly characterized. Unlike conventional receptor-targeted peptide drugs, these compounds have no well-established receptor-agonist mechanism, although receptor-mediated activity has not been excluded. Pinealon is proposed to influence intracellular neuronal stress responses, but the precise molecular targets and causal pathways remain largely unverified. The scientific literature more often describes these peptides through their observed downstream cellular and physiological effects rather than through well-established mechanisms explaining how those effects originate.

Transporter binding and entry are plausible Pinealon mechanisms, supported partly by molecular modeling; predicted LAT1 binding does not itself demonstrate transmembrane transport or human brain exposure. [13] Proposed downstream effects include oxidative-stress regulation, ERK signaling, caspase-related responses, inflammatory mediators, NMDA-receptor gene expression, and dendritic maintenance. [1,4–9]

EDR reduced stress-induced ROS in rat cerebellar granule cells and neutrophil preparations, and reduced necrotic death in stressed PC12 cells. Delayed ERK1/2 activation and altered cell-cycle distribution accompanied these observations. [1] Rat cytokine and caspase findings suggest context-dependent regulation of inflammatory and cell-death pathways. [3,4] Redox effects should not be assumed to be uniformly antioxidant across tissues and exposure conditions. [17]

Biophysical studies demonstrate EDR-DNA interactions in solution; modeling proposes sequence-dependent binding and possible effects on gene regulation. [18,22] Histone binding and RNA-related mechanisms have also been proposed, with evidence varying by experimental system. [8,12] EDR altered NMDA-receptor subunit gene expression in diabetic rats, preserved spines in amyloid-model experiments, and increased dendritic processes and length in induced human neurons. [5–9] These findings support a plausible intracellular mechanism involving gene regulation and neuronal maintenance. They establish observed effects more directly than a single unifying molecular mechanism. Direct gene-regulatory effects in living humans remain plausible.

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

References

[1] Khavinson V, Ribakova Y, Kulebiakin K, et al. Pinealon increases cell viability by suppression of free radical levels and activating proliferative processes. *Rejuvenation Res*. 2011;14(5):535-541. doi:10.1089/rej.2011.1172

[2] Arutjunyan A, Kozina L, Stvolinskiy S, Bulygina Y, Mashkina A, Khavinson V. Pinealon protects the rat offspring from prenatal hyperhomocysteinemia. *Int J Clin Exp Med*. 2012;5(2):179-185. PubMed

[3] Mendzheritskii AM, Karantysh GV, Abramchuk VA, Ryzhak GA. Effect of peptide geroprotectors on navigation learning in rats of different ages and caspase-3 systems in their brain structures. *Adv Gerontol*. 2014;4(1):37–41. doi:10.1134/S2079057014010068

[4] Mendzheritskiĭ AM, Karantysh GV, Ryzhak GA, Dem'ianenko SV. [Regulation of content of cytokines in blood serum and of caspase-3 activity in brains of old rats in model of sharp hypoxic hypoxia with Cortexin and Pinealon]. *Adv Gerontol*. 2014;27(1):94-97. PubMed Russian.

[5] Karantysh GV, Fomenko MP, Menzheritskii AM, Prokof'ev VN, Ryzhak GA, Butenko EV. Effect of Pinealon on Learning and Expression of NMDA Receptor Subunit Genes in the Hippocampus of Rats with Experimental Diabetes. *Neurochem J*. 2020;14(3):314–320. doi:10.1134/S181971242003006X

[6] Kraskovskaya NA, Kukanova EO, Lin'kova NS, Popugaeva EA, Khavinson VK. Tripeptides Restore the Number of Neuronal Spines under Conditions of In Vitro Modeled Alzheimer's Disease. *Bull Exp Biol Med*. 2017;163(4):550-553. doi:10.1007/s10517-017-3847-2

[7] Khavinson V, Ilina A, Kraskovskaya N, et al. Neuroprotective Effects of Tripeptides-Epigenetic Regulators in Mouse Model of Alzheimer's Disease. *Pharmaceuticals (Basel)*. 2021;14(6):515. doi:10.3390/ph14060515

[8] Khavinson V, Linkova N, Kozhevnikova E, Trofimova S. EDR Peptide: Possible Mechanism of Gene Expression and Protein Synthesis Regulation Involved in the Pathogenesis of Alzheimer's Disease. *Molecules*. 2021;26(1):159. doi:10.3390/molecules26010159

[9] Kraskovskaya N, Linkova N, Sakhenberg E, et al. Short Peptides Protect Fibroblast-Derived Induced Neurons from Age-Related Changes. *Int J Mol Sci*. 2024;25(21):11363. doi:10.3390/ijms252111363

[10] Umnov RS, Lin'kova NS, Khavinson VKh. [Neuroprotective effects of peptides bioregulators in people of various age]. *Adv Gerontol*. 2013;26(4):671-678. PubMed Russian.

[11] Bashkireva AS, Artamonova VG. [The peptide correction of neurotic disorders among professional truck-drivers]. *Adv Gerontol*. 2012;25(4):718-728. PubMed Russian.

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

[12] Khavinson VK, Popovich IG, Linkova NS, Mironova ES, Ilina AR. Peptide Regulation of Gene Expression: A Systematic Review. *Molecules*. 2021;26(22):7053. doi:10.3390/molecules26227053

[13] Khavinson V, Linkova N, Kozhevnikova E, Dyatlova A, Petukhov M. Transport of Biologically Active Ultrashort Peptides Using POT and LAT Carriers. *Int J Mol Sci*. 2022;23(14):7733. doi:10.3390/ijms23147733

[14] US Food and Drug Administration. Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers. Published July 2005. FDA guidance

[15] Khavinson V, Ilina A, Kraskovskaya N, et al. Correction: Khavinson et al. Neuroprotective Effects of Tripeptides-Epigenetic Regulators in Mouse Model of Alzheimer's Disease. Pharmaceuticals 2021, 14, 515. *Pharmaceuticals (Basel)*. 2025;18(1):111. doi:10.3390/ph18010111

[16] Khavinson VK, Grigoriev EI, Malinin VV, Ryzhak GA, inventors. Peptide substance stimulating regeneration of central nervous system neurons, pharmaceutical composition on its base, and the method of its application. US patent application US20090105157A1. Published April 23, 2009. Patent record

[17] Meshchaninov VN, Tkachenko EL, Zharkov SV, Gavrilov IV, Katyreva IuE. [EFFECT OF SYNTHETIC PEPTIDES ON AGING OF PATIENTS WITH CHRONIC POLYMORBIDITY AND ORGANIC BRAIN SYNDROME OF THE CENTRAL NERVOUS SYSTEM IN REMISSION]. *Adv Gerontol*. 2015;28(1):62-67. PubMed Russian.

[18] Khavinson V, Linkova N, Kukanova E, et al. Neuroprotective Effect of EDR Peptide in Mouse Model of Huntington's Disease. *J Neurol Neurosci*. 2017;8(1):166. doi:10.21767/2171-6625.1000166

[19] Marín-Jerez E, Rueda-Carrasco J, Meléndez-Rodríguez F, et al. Short-Term Performance Assay Identifies Functional Benefits and Early Toxicity of Longevity Interventions in Mice. *bioRxiv*. Preprint posted February 26, 2026. doi:10.64898/2026.02.25.707674

[20] Nazimko VA, Morgul' EV, Petrova OA, et al. [Analysis of some parameters of biological age and adaptation possibilities of workers of locomotive brigades]. *Adv Gerontol*. 2012;25(1):57-62. PubMed Russian.

[21] Khavinson VKh, Lin'kova NS, Tarnovskaya SI, Umnov RS, Elashkina EV, Durnova AO. Short peptides stimulate serotonin expression in cells of brain cortex. *Bull Exp Biol Med*. 2014;157(1):77-80. doi:10.1007/s10517-014-2496-y

[22] Silanteva IA, Komolkin AV, Morozova EA, Vorontsov-Velyaminov PN, Kasyanenko NA. Role of Mono- and Divalent Ions in Peptide Glu-Asp-Arg-DNA Interaction. *J Phys Chem B*. 2019;123(9):1896-1902. doi:10.1021/acs.jpcb.8b10359

[23] Myakotnykh VS, Torgashov MN, Egorin KV, et al. [Comparative analysis of different methods of geroprotective]. *Adv Gerontol*. 2016;29(4):594-601. PubMed Russian.